Evaluating Dynamic Constraint in Belief Systems*

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Do political belief systems exhibit dynamic constraint? Drawing on a bounded rationality account of belief formation, I examine whether coordinated belief change occurs when credible information and cross-issue cues target a range of political beliefs. I test this expectation in a pre-registered survey experiment in the United States ($N = 2{,}414$), in which participants are presented with factual information about income inequality and social mobility—with and without explicit target cues, which indicate that the two issues are conceptually linked. The information treatments produced large first-stage effects on their respective target beliefs, while leaving peripheral beliefs unaffected. In exploratory analyses, I extend these analyses to economic preferences and people's self-perceptions of belief consistency. The findings delimit the scope of dynamic constraint and suggest that belief updating may be largely domain-specific.

Disclaimer: This is no longer a working paper, and the article will not be revised or submitted for publication. I provide the final version of the article below for scientific record. The data and code is presented at github.com/tkeskinturk/dynamic_constraint

Introduction

Studies of political belief systems propose that many citizens hold beliefs and preferences that "go together" at a single point in time—what Converse (1964) calls *static constraint* (see Boutyline and Vaisey 2017; Brandt and Sleegers 2021; Fishman and Davis 2021). However, it is far less common for people to exhibit *dynamic constraint*, the tendency for a change in one belief to induce congruent shifts in logically or sociologically related beliefs. While there are strong cognitive accounts of why people might seek consistent belief states (Cartwright and Harary 1956; Heider 1946; Osgood and Tannenbaum 1955), the experimental evidence suggests that dynamic constraint remains weak in

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both mass and elite populations (Coppock and Green 2022; Turner-Zwinkels and Brandt 2022).

One promising explanation attributes this pattern to cognitive intractability. When presented with information, people may update the focal belief that the new information makes salient, but rarely retrieve or reconcile distal considerations unless prompted to do so (Sommer, Musolino, and Hemmer 2023). Because a full search of memory is often unfeasible—and potentially highly complex unless triggered by political elites (Gonthier and Guerra 2022; Groenendyk, Kimbrough, and Pickup 2023; Keskintürk 2022b) or cognitive authorities (Rawlings 2020)—people tend to rely on whatever beliefs are most accessible at any time. Hence, when confronted with new information, people may only allocate scarce cognitive resources to "constrain" their belief networks (e.g., Simon 1955).

This perspective implies that it may be possible to observe dynamic updating *if people could observe connections between issues explicitly*. Suppose that an actor has a focal belief, ρ_i, as well as a peripheral belief, ρ_j.[1] If this actor's beliefs were fully tractable, they would prefer a cognitively consistent state; hence, there is a penalty $\lambda > 0$ from inconsistent belief states (i.e., $\rho_i - \rho_j \neq 0$). Suppose that there is an action set, $a \in \{0, 1\}$, where 1 denotes a consistency-seeking belief system search and 0 denotes no action. This means that if an actor engages in $a = 1$, the search will move ρ_j to $\tilde{\rho}_j = \rho_i$, eliminating dissonance. That said, belief system search comes with a search cost, k, for the actor.

Suppose now that this cost might be lowered with exogenous cues $c \in [0, 1]$, where $c = 0$ means no cue from the environment while $c = 1$ is a clear and trustworthy signal. In this setting, mechanisms like elite framing (Zaller 1992) or associative learning (Goldberg and Stein 2018) serve as cognitive cost reduction processes. Hence, the utility model captures a fundamental trade-off:

$$U(\rho_j, a) = -(1 - a)\lambda(\rho_i - \rho_j)^2 - ak(1 - c) \tag{1}$$

The final decision rule is simple: when $\lambda(\rho_i - \rho_j)^2 < k(1 - c)$, the actor has no incentive to engage in consistency-seeking belief system search, which likely characterizes most experimental settings. In Figure 1, I show the implications of this simple model on choice behavior across c and $\rho_i - \rho_j$.

The empirical prediction from this model is that if there is a cue—i.e., higher c—the actor can spend

[1] I assume that ρ_i and ρ_j are (a) continuous, (b) bounded between 0 and 1, and (c) similar in ideological valence.

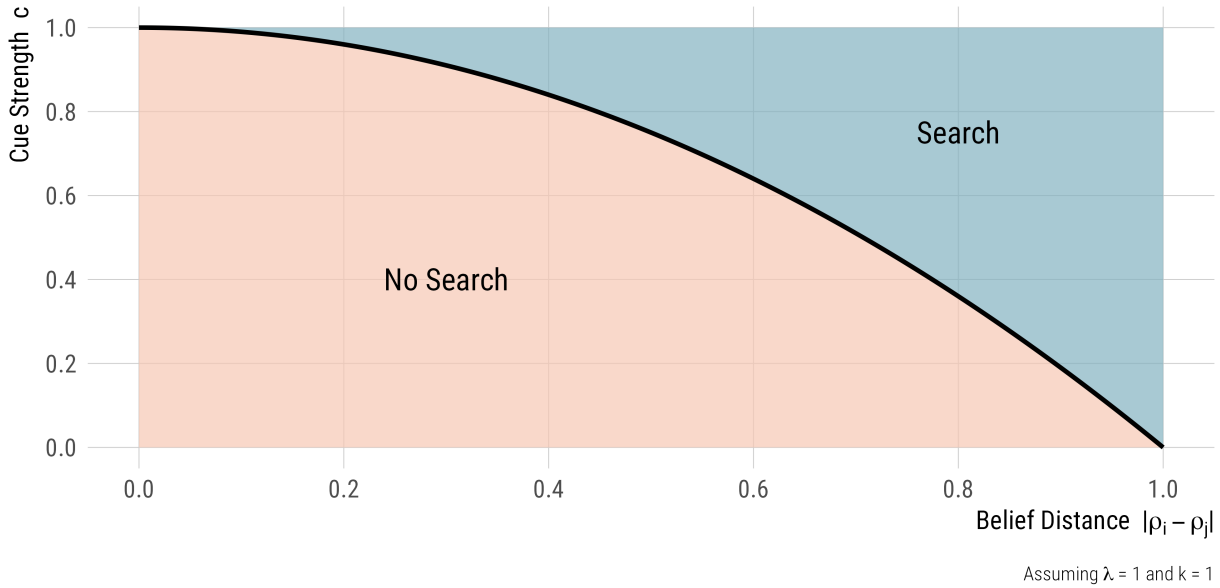


Figure 1: Consistency-Seeking Decision Rule Adjudicating Search or No Search

fewer resources to update beliefs. The corresponding generative model for this prediction is rather simple: when presented with information, a person will update their focal belief, but the distal or peripheral belief will be dynamically updated only when there is a cue that connects the focal belief to the peripheral belief. If memory search is indeed costly, cues that tie one issue consideration to another can thus provide a low-cost alternative for achieving consistent or congruent beliefs. In this sense, dynamic updating may occur when the information environment actively supplies cues that make the otherwise latent link between issue considerations explicit, as this would lower retrieval costs for a full belief system search process and invite coordinated and dynamic belief updating.

The coherence of this theoretical framework relies on reasonable updating rules among individuals when faced with new information. Therefore, a strictly empirical first expectation proposes that

> Hypothesis 1: When an individual encounters credible factual information about a focal issue, they should update their beliefs in the same direction as the evidence.

The model notes that, without cues, cross-issue updating is less likely. Of course, the inconsistency penalty λ may still produce updating without cues if the penalty is sufficiently high for the person. That said, the probability of these cross-issue updates in contexts where there is no exogenous cues

should still be drastically lower. At the same time, it is also plausible to think that the penalty term is arguably lower in an experimental context. These considerations imply that

> Hypothesis 2: Since belief system search is usually costly, updating should not extend to conceptually related but unmentioned issues in the absence of cues.

Similarly, the same logic implies that

> Hypothesis 3: Providing people with explicit cues linking the focal issue to a peripheral issue should result in cross-issue updating, generating coordinated belief change.

Together, these three hypotheses provide a powerful theoretical test of whether making latent issue connections salient among the population might generate dynamic constraint.

The Empirical Setting

I test these propositions in a pre-registered survey experiment that builds on Coppock and Green's (2022) information design. I randomly assign participants (N = 2,414) to view factual information about either income inequality or social mobility in the United States, with and without an explicit cue indicating that the two issues are conceptually linked, resulting in a five-arm experimental setting. While the treatments target focal beliefs within their respective domains—aiming to generate strong first-stage effects—the interest lies in whether these effects would produce spillover effects in peripheral domains, or put differently, respective second-stage effects.[2]

Beliefs About Inequality and Social Mobility

Over the past several decades, income inequality in the U.S. has increased markedly (Piketty and Saez 2003) while intergenerational mobility has weakened (Chetty et al. 2017). Studies often show that beliefs about inequality and mobility are frequently confounded in public opinion, with perceptions of mobility moderating how concerned the public is about inequality (Heiserman, Simpson, and Willer 2020). Several theoretical models argue that high levels of income inequality might

[2]The experimental protocol and the main hypotheses are pre-registered, as presented in https://osf.io/u49av.

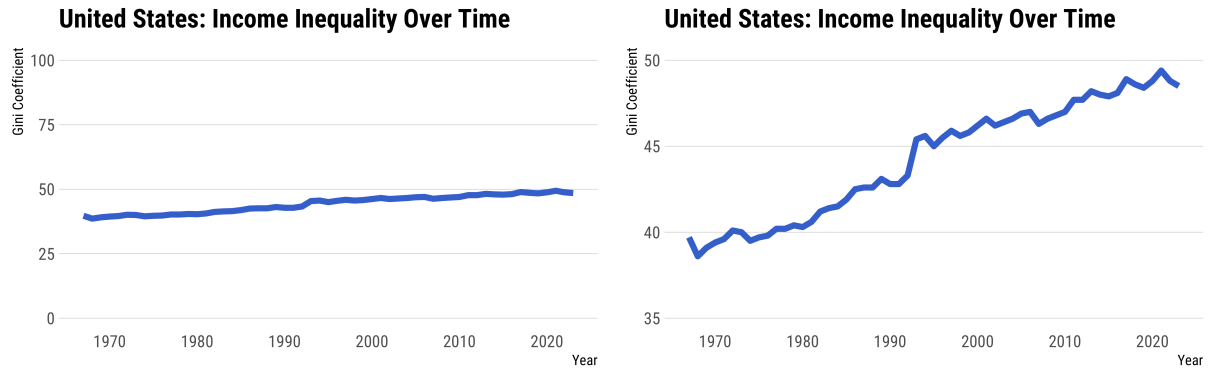


Figure 2: Inequality Treatment

Notes: The figure presents the inequality treatment. The left panel shows the Gini coefficient values as presented to the control group. The right panel shows the Gini coefficient values as presented to the treatment group. In both plots, the underlying data is exactly the same, while the y-axes are manipulated via graph scaling.

be reframed as acceptable for the public when paired with the belief that mobility is also high, consistent with the notion that opportunity legitimizes unequal outcomes (Bénabou and Tirole 2006). In that sense, many Americans hold a tightly coupled worldview in which awareness of inequality coexists with confidence in mobility, sustained by a broader belief in meritocracy.

I bring these domains together in a shared design, where the goal is to manipulate the perceptions of one domain to see its effects on the other domain. In doing so, I incorporate two treatments from previous research—one about income inequality and one about social mobility—while holding the other constant, so that any cross-issue effects can be attributed to belief system search rather than joint information exposure. Figure 2 presents the inequality treatment, adapted from Trump and White (2018), where participants observe a flat (control group) or a strongly increasing (treatment group) trajectory of the Gini coefficient in the U.S. through a graph-scaling manipulation. Figure 3 presents the mobility treatment, adapted from Alesina, Stantcheva, and Teso (2018), which shows an informative animation relying on expert knowledge about U.S. social mobility.

Cues and Consistency-Seeking

To examine whether a conceptual link between issues—i.e., a higher c—might trigger coordinated updating, I implement two additional conditions ("cue conditions") that append a short relational

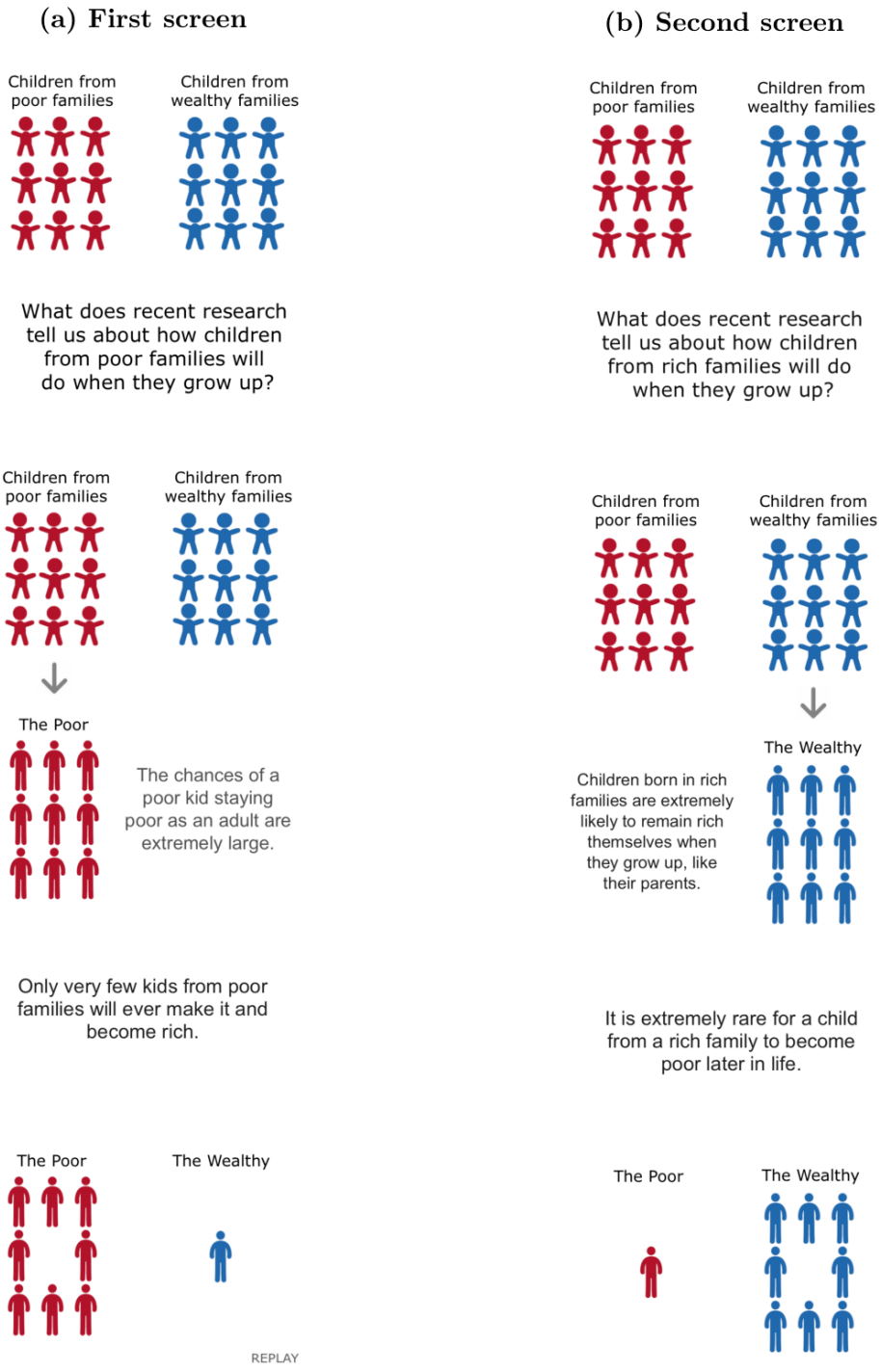


Figure 3: Social Mobility Treatment

statement to the original treatments. These statements balance a theoretical trade-off: stronger cues that explicitly tie the two domains risk functioning as a *secondary treatment* through generating joint

information exposure, while weaker cues may fail to activate cross-issue retrieval. Hence, I opted to keep the connections minimal, while at the same time explicitly noting the link between issues.

In what I call the "Inequality + Cue Condition," participants first saw the treatment presented in Figure 2, immediately followed by a standalone cue that states:

> *Cue for Inequality Condition*: "Some experts suggest that income inequality may be linked to differences in social mobility—the ability to move up or down the economic ladder."

For the "Social Mobility + Cue condition," the cue following Figure 3 noted:

> *Cue for Social Mobility Condition*: "Some experts suggest that limited social mobility may be connected to income inequality over time."

Once again, the expectation is that, without these cues, cross-issue updating should be non-existent or minimal (H2), while with these cues, cross-issue updating should be present or higher (H3).

Data and the Experimental Setting

To evaluate these hypotheses, I recruited 2,648 participants from *Prolific*, an online survey platform, in April 2025. I targeted a convenience sample of English-speaking U.S. adults, with quotas on sex, age, and political affiliation. Following exclusions, the final sample consisted of 2,414 participants. With five treatment arms (explained below), this resulted in 480–486 participants in each condition, which gave me more than 85% power to detect an effect of $d = 0.20$ in a two sample test.

In the survey questionnaire, participants answered blocks of questions assessing their beliefs about inequality, social mobility, and economic preferences. Inequality beliefs included questions about the rise in income inequality, agreement that there is too much inequality in the U.S., and attributions for poverty to lack of effort. Mobility beliefs included questions about the chances of a poor child reaching the upper class, the opportunities for being successful in the U.S., and the influence of parental status on adult socio-economic status. I followed these beliefs by collecting participants' opinions on six economic policies, including redistribution, poverty alleviation, progressive taxation, education and job-training funding, early-childhood programs, and inheritance taxation.

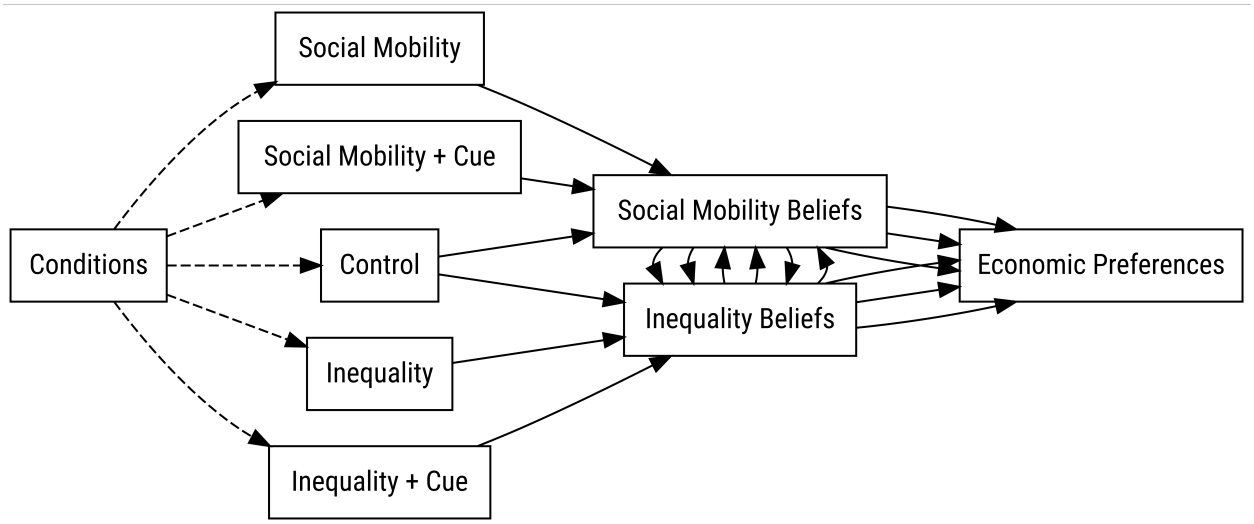


Figure 4: The Flowchart for the Survey Protocol

Figure 4 shows the basic survey protocol and the order in which all question blocks were presented. I randomly assigned participants to one of five conditions: a control group, two information treatments (*inequality* or *mobility*) and two cue conditions (*inequality + cue* or *mobility + cue*). After each treatment, participants first completed the block of questions about their focal issues, followed by the peripheral issues, and ending up with the economic preference measures. Before the end of the questionnaire, I presented all participants with their responses to the inequality and social mobility questions, asking whether they think their responses were consistent or inconsistent.

Supplemental Materials A provides more details about the survey protocol, questionnaire, quality controls, and missing values. Supplemental Materials B provides information about balance checks across conditions, the distribution of beliefs and preferences in the sample, and a check on question order effects, which rules out order effects as a potential explanation for the following results.

Findings

The analyses below show that the information treatments substantively change beliefs about their target issues (confirming H1), while leaving non-target beliefs unchanged in both the no-cue and

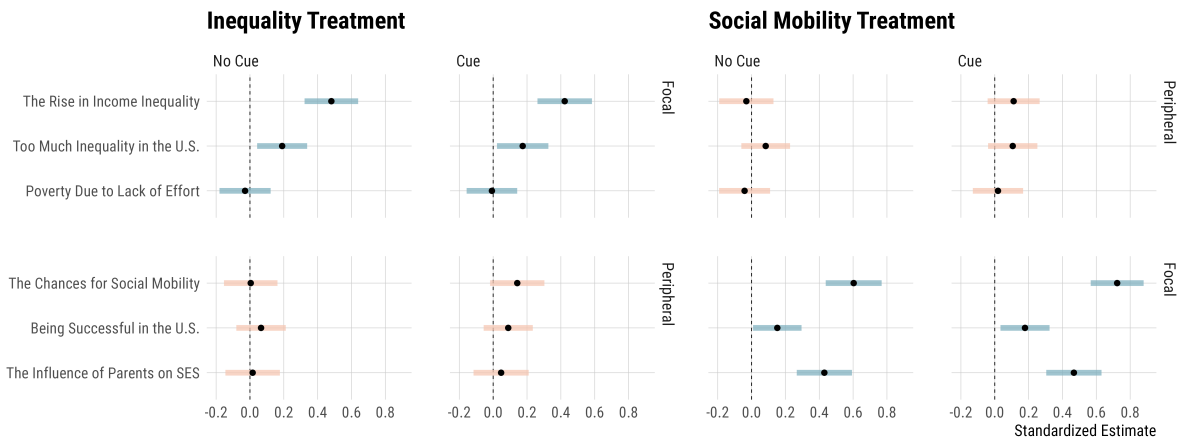


Figure 5: Treatment Effects on Inequality and Mobility Beliefs

Notes: The figure presents the treatment effects of the inequality treatment and the social mobility treatment on both the inequality and social mobility beliefs, classified across no-cue and cue conditions. 95% confidence intervals are corrected via Bonferroni correction for multiple comparisons. Outcomes are standardized to have a mean of 0 and an SD of 1.

cue conditions, which implies that while H2 is confirmed, the main prediction (H3) is rejected.

In Figure 5, we observe that the focal beliefs (indexed as blue items) update in the treatment groups compared to the control group. These updates occur both in the no-cue (Conditions 2 and 3) and cue (Conditions 4 and 5) conditions, with roughly similar estimates. When it comes to the spillover of these treatments to peripheral issues, however, we see that neither the no-cue condition nor the cue condition can actually produce a spillover effect. While the effects are in the expected direction in the cue conditions, these are statistically insignificant and highly imprecise.[3]

In exploratory analyses, I look at the spillover effects from beliefs to preferences. Since all six preferences correlated strongly (with a Cronbach's alpha of 0.88 and a single factor loading), I created an index of economic preferences for this analysis.[4] In the first set of models, I first regressed this index on *the rise in income inequality* belief from the inequality block and *the chances for social mobility* belief from the social mobility block, in separate models, adjusting for various covariates.[5] In the

[3] In the pre-registration, I predicted that it may be possible that "political information and partisanship may moderate these effects." In exploratory analyses, I interacted treatment status with political attention and party identity for the central outcome variables ("the rise in income inequality" and "the chances for social mobility"), and compared the Bayesian Information Criteria (BIC) between the interaction models and the main models. I found that none of the interactions decreased the BIC, suggesting no model improvement with these additional analyses.

[4] Doing these analyses separately for each question leads to the same findings.

[5] The adjustment set includes age, gender, race, education, income, party affiliation, political ideology, and self-reported attention to politics and news.

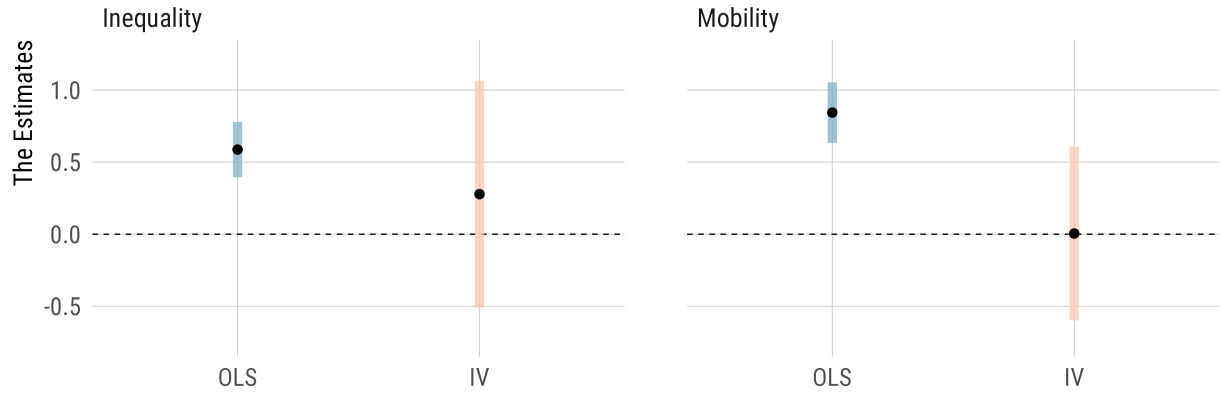


Figure 6: Treatment Effects on Economic Preferences

Notes: The figure presents the effects of inequality and social mobility beliefs on a standardized index of economic preferences. The OLS estimates show the simple OLS coefficients for respective beliefs on preferences, adjusting for covariates. The IV estimates show the effects of each belief on preferences, where beliefs are instrumented by the experimental conditions using a two-stage least squares procedure. All estimates show robust standard errors.

second set of models, I used the treatment variable as an instrumental variable to exploit the exogenous variation in inequality and mobility beliefs to estimate their subsequent effects on preferences. Figure 6 shows these results. Once again, there is no evidence of dynamic constraint.

How do participants see their own beliefs? In a final set of exploratory analyses, I showed respondents their answers on *the rise in income inequality* and *the chances for social mobility*, and asked them whether they think that their responses to these questions are "consistent." As seen in the left panel of Figure 7, the majority of participants reported that their responses are indeed consistent, but a sizable set (roughly 30%) sees their responses somewhere in between. Looking at the correlations between these two measures across groups, I indeed saw that these beliefs have a polychoric correlation of 0.30 among those who state that they are consistent, while the correlation is -0.06 among those who state that they are in-between or inconsistent in their response patterns.

Discussion

In this article, I evaluated whether dynamic constraint may emerge when credible information and cross-issue cues target political beliefs. Using a survey experiment in the United States, I found that

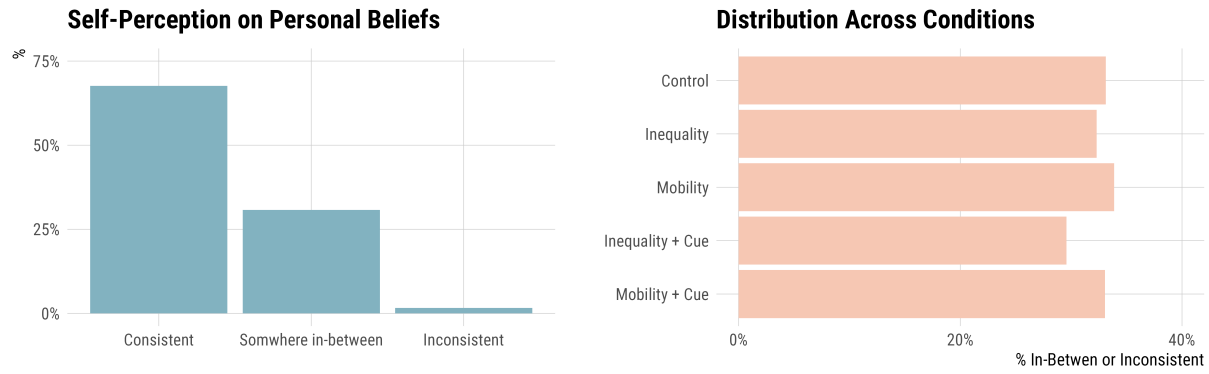


Figure 7: The Distribution of Self-Reports on Consistency

Notes: The figure presents the distribution of self-reports on response consistency. The left panel shows the percentage of participants indicating that their responses to the inequality and mobility beliefs are consistent or inconsistent. The right panel shows the percentage of somewhere-in-between or inconsistent participants across experimental conditions.

while information treatments reliably generate strong first-stage effects, the spillover effects do not occur in either no-cue or cue environments. These findings further delimit the scope for the theory of dynamic constraint: merely making an inter-issue connection explicit is insufficient to produce coordinated belief change. More importantly, if cues fail to produce belief change, this raises doubts about whether dynamic constraint is a mechanism of belief formation in the first place.

One interpretation of these findings is that the correlations in static belief systems may simply be a reflection of long-term socialization and ideological identity (Boutyline and Vaisey 2017) than the result of cumulative consistency-seeking belief updates (Goldberg and Stein 2018). This finding is thus consistent with the possibility that belief coherence emerges over the life-course, via gradual accumulation of linked considerations (Keskintürk 2022a) instead of "active updating."

Of course, the experimental design as executed in this article—brief information exposure, minimal stakes, a rather small cue, and the limited issue set—might underestimate the effect of cues in more sustained and politically charged contexts. The next step is to examine whether dynamic constraint emerges under conditions of prolonged exposure, elite endorsement, or higher personal relevance. Ultimately, clarifying whether belief constraint is an underlying cause or a mere reflection of belief system structure is imperative for building a generative theory of political opinionation.

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Supplemental Materials for

Evaluating Dynamic Constraint in Belief Systems

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March, 2026

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1 Introduction

In this document, I provide the supplemental materials for the paper, *Evaluating Dynamic Constraint in Belief Systems*. The research was approved by the Institutional Review Board at Duke University, with the protocol code 2025-0397: *Evaluating Dynamic Constraint in Political Belief Systems*.

The replication files include the complete source files to reproduce the findings of this article, and the QSF file that can reconstruct the survey. They are stored in the author's GitHub page.

2 Supplemental Materials A: Data Collection

I conducted the experiment in *Prolific* using a custom "Qualtrics" survey. In this section, I document the details regarding the survey protocol, questionnaire, and quality controls.

2.1 Survey Protocol

I recruited 2,648 survey participants from the online survey platform *Prolific* in April 2025. Of these, 98 participants did not return their surveys or were excluded due to an initial attention check.[1] I excluded an additional 114 participants as Qualtrics flagged them as likely bots or likely duplicates, as well as several participants failing a pre-registered attention check that flagged them if their self-reported age and year of birth had a difference of 2 or more years.[2] Excluding these participants, as well as 22 with missing values, reduced the final analytical sample to 2,414.

I set U.S. representative quotas across sex, age, and political affiliation.[3] Participants then selected into a survey titled "Economic Issues of the Day," with the description saying:

> *"In this study, we are interested in people's beliefs and preferences on several economic issues of the day. The survey takes roughly 6 minutes to complete."*

I allowed both mobile and non-mobile access to the survey, and participants were paid $1 for their participation. The median survey completion time was roughly 7 minutes. Looking at the patterns, older people, Blacks and Hispanics took slightly longer to finish the instrument.

2.2 Questionnaire

The survey document was roughly organized into seven distinct blocks:

[1] Due to an IRB request, I had to exclude these participants from the source files as they did not get paid.

[2] I interacted the treatment status with a quality flag for the 114 participants, finding no substantive differences in how treatments relate to the outcomes. Hence, I dropped these participants from consideration.

[3] Unfortunately, Prolific did not allow me to include race, as including it dramatically reduced the number of available participants in its pool. Hence, the trade-off was between adequate race representation and political representation, and due to the nature of the questionnaire as well as the study aims, I opted for political representation.

(1) An introductory section, where participants went through several security checks, as detailed in "Quality Controls" below, and provided their consent to the research.
(2) A warm-up section, where participants provided information about their age, political knowledge[4], attention to politics, ideology, partisan identity, and year of birth.[5]
(3) The treatment blocks, where participants were assigned to a control condition or one of four treatment conditions.
(4) A section on questions about economic inequality and social mobility.
(5) A section on questions about economic preferences.
(6) A section on response consistency and open-ended responses probing that consistency.
(7) A closing section with questions on gender, race, education, and income.

In the consent form, I tried my best to mitigate concerns about the survey, saying:

> *"We are a non-partisan group of academic researchers from Duke. Our goal is to understand the political beliefs and preferences of people in the U.S. No matter what your political views are, this is an important question, and you are contributing to our knowledge as a society. You might not agree with all the information presented, and that is perfectly fine. Our survey will give you an opportunity to express your own views."*[6]

As described in the main manuscript, participants were assigned to one of five conditions: a control condition, two information conditions, and two cue conditions. For the information conditions, I forced participants to remain on the page for 10 seconds for the inequality treatment and 17 seconds for the mobility treatment. For the cue condition, participants were forced to wait an additional 8 seconds on the cue page, in addition to their wait times in the information page.

The *Question Block 1* contained questions on inequality, randomized within the page:

- Compared to 30 years ago, the income inequality in the United States is:
  much lower, *lower*, *about the same*, *higher*, *much higher*.

[4]The percent of participants correctly answering these questions was very high; hence, I do not use these items.
[5]I presented the age as the first question and year of birth as the last question within the same block.
[6]The rest of the consent form included a standard briefing on data privacy and information on researchers and IRB.

- Differences in income in the United States are too large:
  strongly disagree, *somewhat disagree*, *neither agree nor disagree*, *somewhat agree*, and *strongly agree*.
- If a person is poor, that's due to a lack of effort on his or her part, rather than circumstances beyond his or her control:
  strongly disagree, *somewhat disagree*, *neither agree nor disagree*, *somewhat agree*, and *strongly agree*.

The *Question Block 2* contained questions on social mobility, randomized within the page:

- The chances that a child from the poorest families in the U.S. will grow up to be among the upper-class families are:
  very low, *fairly low*, *moderate*, *fairly high*, *very high*.
- In the United States, everybody has a chance to make it and be economically successful:
  strongly disagree, *somewhat disagree*, *neither agree nor disagree*, *somewhat agree*, and *strongly agree*.
- Most people's economic positions in adulthood are strongly influenced by their parents' economic status:
  strongly disagree, *somewhat disagree*, *neither agree nor disagree*, *somewhat agree*, and *strongly agree*.

The *Question Block 3* contained questions on policy preferences, with no randomization:

- (Adapted from the *GSS*): Some people think that the government in Washington ought to reduce the income differences between the rich and the poor, perhaps by raising the taxes of wealthy families or by giving income assistance to the poor. Others think that the government should not concern itself with reducing this income difference between the rich and the poor. What score between 1 and 7 comes closest to the way you feel?
  1: *the government should reduce income differences*,
  7: *the government should not concern itself with reducing income differences.*
- (Adapted from the *GSS*): Some people think that the government in Washington should do everything possible to improve the standard of living of all poor Americans. Other people think it is not the government's responsibility, and that each person should take care of himself. What score between 1 and 7 comes closest to the way you feel?

1: *the government should improve the living standards of the poor*,

7: *each person should take care of himself or herself.*

- How much do you support or oppose making the federal income tax more progressive—for example, adding higher tax brackets for very high incomes?
  strongly oppose, *oppose*, *slightly oppose*, *neutral*, *slightly favor*, *favor*, *strongly favor*.
- How much do you support or oppose increasing federal funding for job training and higher education grants to help low-income Americans move up economically?
  strongly oppose, *oppose*, *slightly oppose*, *neutral*, *slightly favor*, *favor*, *strongly favor*.
- Would you favor or oppose new government programs—such as universal pre-K or subsidized childcare—aimed at giving low-income children better early opportunities?
  strongly oppose, *oppose*, *slightly oppose*, *neutral*, *slightly favor*, *favor*, *strongly favor*.
- Inheritance taxes should be increased to prevent wealth from unfairly determining people's opportunities in life.
  strongly oppose, *oppose*, *slightly oppose*, *neutral*, *slightly favor*, *favor*, *strongly favor*.

In the open-ended section of the instrument, I provided two questions for the participant. The first question retrieved participant's answers to the inequality and social mobility questions, saying:

> *Previously, you stated that, compared to 30 years ago, the income inequality in the United States is {X} and the chances that a child from the poorest families in the U.S. will grow up to be among the upper-class families are {Y}. People often hold different beliefs about income inequality and social mobility without explicitly considering how these beliefs fit together.*
>
> *After seeing your previous answers, do you feel your beliefs are consistent, inconsistent, or somewhere in between?*

Immediately after this question, participants answered an open-ended text question that retrieved their answer from above and asked why they think their answer was consistent or inconsistent.

2.3 Quality Controls

I implemented several security and qualification checks before, during, and after the survey.

In the Qualtrics instrument, I embedded two qualification checks. First, I collected participant IPs and sent them to IPHub Proxy, an IP lookup website featuring proxy and VPN detection, preventing participants with IPs detected as proxies or those using VPNs from continuing with the survey. Second, I used IPHub country parameters to ensure participation from within the U.S.

As noted before, I conducted two attention checks. In the warm-up, I asked participants, "For our research, careful attention is critical! To show that you are paying attention, please select <I have a question.>" Those who answered this question incorrectly were immediately excluded from the survey. Second, as pre-registered, I used the difference between participants' age and year of birth as a second qualification—excluding those whose answers differed by 2 or more years.

The built-in Qualtrics assessments served as the final quality control, where I screened out participants who were flagged as likely bots or duplicates from the final sample of participants.

2.4 Missing Values

After pre-processing, there were 22 participants with missing values in the covariates. These values were all in the final closing section, with 13 people having missing data on education, 14 on income, 3 on party affiliation, and 6 on political ideology. In the pre-registration, I noted that:

> **Note on Missing Data**: *If the participant has missing data on outcomes, that participant will not be included in the analyses. If the participant has missing data on controls, that participant will be included in all the analyses, with their covariance* [*sic*] *value set to an imputed value.*

Since there were very few participants with missing data, and their inclusion did not have an effect on any of the analyses, I decided to deviate from the plan and drop these participants.

3 Supplemental Materials B: Descriptive Information

In this section, I will present a variety of descriptive information about the sample and the measures employed, the distribution of outcomes, as well as analyses on question order effects.

3.1 Balance Checks

While preparing the data for analyses, I made several decisions in coding the covariates. I collapsed all responses to the gender question to *male* and *female*[7], organized race into four categories (*white*, *black*, *Hispanic*, and *other race*), organized education similarly into four categories (*high school or less*, *some college*, *college*, and *post-graduate*), and calculated the midpoints of the bracketed responses for income (e.g., assigning $\$17,499.5$ for the category $\$15,000 - \$19,999$). Additionally, I normalized political attention and ideology measures to range between 0 and 1, and combined political affiliation into Democrat, Republican, and Independent; the last category including both those reporting that they are Independent, and those saying that they follow another party.

With these notes, Table B1 presents a balance table for all covariates across conditions, indicating that the conditions are, on observed characteristics, appropriately balanced.

Table B1: Descriptive Statistics Across Conditions

Characteristic	**C** N = 483	**T1** N = 486	**T2** N = 484	**T3** N = 480	**T4** N = 481
Age	44.29	44.09	46.34	45.29	44.96
Gender					
Male	48.0%	50.0%	47.1%	50.2%	49.1%
Female	52.0%	50.0%	52.9%	49.8%	50.9%
Race					
White	74.9%	70.6%	74.0%	75.4%	73.4%
Black	11.4%	13.6%	11.6%	11.0%	12.3%
Hispanic	4.8%	6.2%	6.6%	4.6%	5.4%
Other	8.9%	9.7%	7.9%	9.0%	8.9%

[7]The original questionnaire included response options *non-binary / third gender* and *prefer not to say*.

Educational Attainment					
HS or Less	15.9%	12.6%	17.6%	20.0%	17.7%
Some College	25.5%	24.3%	22.1%	22.5%	20.2%
College	44.3%	45.3%	44.0%	40.4%	44.1%
Post-Graduate	14.3%	17.9%	16.3%	17.1%	18.1%
Income ($)	76,805.93	81,378.13	76,523.29	75,218.27	76,605.55
Party Identification					
Democrat	33.7%	35.0%	33.7%	34.8%	33.7%
Independent	40.6%	38.1%	35.1%	39.4%	35.8%
Republican	25.7%	27.0%	31.2%	25.8%	30.6%
Ideological Identification	0.45	0.45	0.48	0.45	0.46
Political Attention	0.66	0.64	0.65	0.65	0.65

Notes: C is the control condition, T1 is the inequality condition, T2 is the mobility condition, T3 is the inequality cue condition, and T4 is the mobility cue condition.

3.2 Descriptive Statistics on Beliefs and Preferences

In Figures B1 and B2, I present the distribution of outcome measures across all conditions. To make the comparisons clear, all responses are coded such that higher numbers indicate liberal views.

There are two important implications of these plots. First, while ideological identity correlates very strongly both with "the rise in income inequality" ($\rho = 0.30$) and "the chances for social mobility" ($\rho = 0.51$)—two central measures of descriptive beliefs—it is important to note that there are many individuals believing that inequality in the U.S. has increased and the chances for social mobility are relatively low. Hence, to the extent that information treatments might have affected individual beliefs, the window of variation is rather narrow. Second, economic preferences in Figure B2 show, in general, two peaks, having clear connections to ideological identity (correlations with ideology range from $\rho = 0.47$ to $\rho = 0.60$), suggesting constrained preferences.

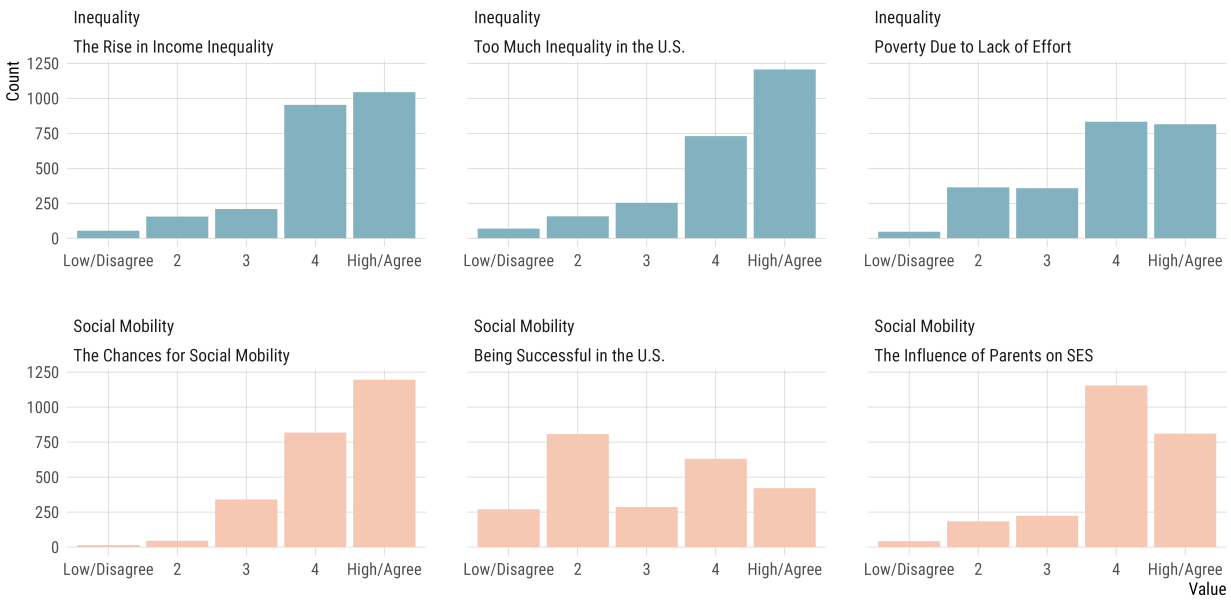


Figure B1: The Distribution of Inequality and Mobility Outcomes

Notes: The figure shows bar plots describing the distribution of responses across belief items on inequality and mobility measures. When appropriate, responses are recoded such that higher values represent more liberal positions.

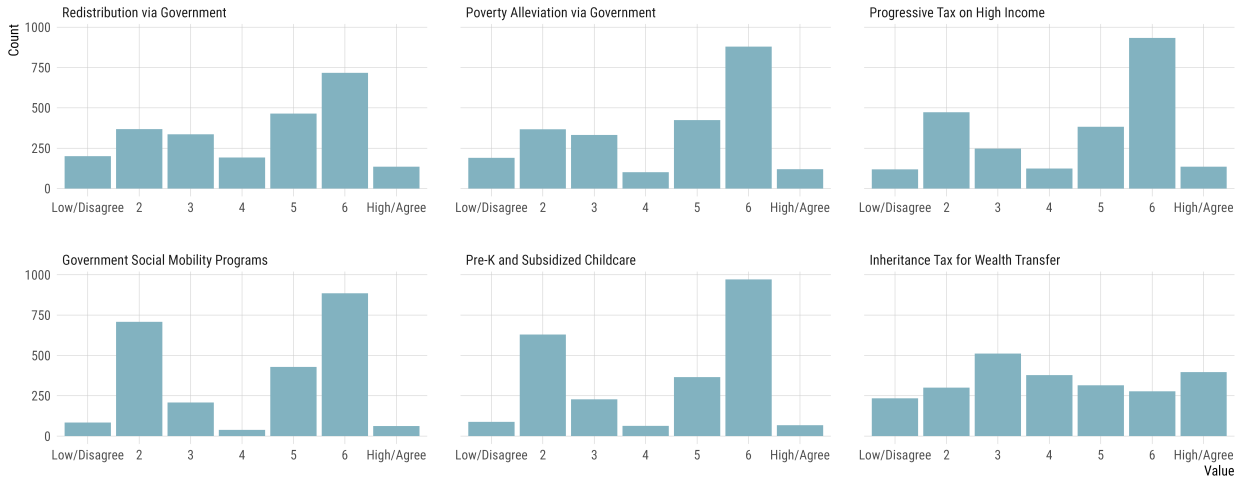


Figure B2: The Distribution of Economic Preferences

Notes: The figure shows bar plots describing the distribution of responses across preference items on economic issues. When appropriate, responses are recoded such that higher values represent more liberal positions.

3.3 Question Order Effects

One potential problem in the structure of the questionnaire is that, since question blocks follow the treatment conditions immediately, the responses may be contaminated by ordering effects. To as-

sess whether this is a serious concern, I randomized the order of inequality question block and the mobility question block within the control group, such that $N = 236$ of the participants answered the inequality questions first and the mobility questions second, and $N = 247$ of the participants answered the mobility questions first and the inequality questions second.

Table B2 shows OLS estimates of outcomes on a question order dummy, showing that there are no significant differences between the two groups, ruling out that possibility.

Table B2: Question Order Effects

	(1)	(2)	(3)	(4)	(5)	(6)
Order	0.01	0.01	-0.02	0.03	0.05	-0.00
	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.02)
N	483	483	483	483	483	483
R2	0.00	0.00	0.00	0.01	0.01	0.00

Notes: OLS estimates of order effects within the control group. Model 1 analyzes the rise in income inequality. Model 2 analyzes too much inequality in the U.S. Model 3 analyzes poverty due to lack of effort. Model 4 analyzes the chances for social mobility. Model 5 analyzes being successful in the U.S. Model 6 analyzes the influence of parents on SES. Robust standard errors in parantheses.